

ReCodeAgent / CodeWeaver Reproducibility Report

Completion Verdict

Status: COMPLETE
Coverage fraction (raw_runs / full requested matrix): 1.0
All completion criteria met.

Manifest

118 / 118 projects discovered.

Execution Coverage (raw_runs, by variant x tool)

variant	tool	expected	measured	coverage_fraction
full	alphatrans	12	12	1.0000
full	crust	300	300	1.0000
full	oxidizer	18	18	1.0000
full	skel	24	24	1.0000

Blockers (collect.py failures.csv)

No failures recorded (or failures.csv was not supplied).

Blockers (test_compare.py comparison_failures.csv)

No failures recorded (or test_compare.py has not been run / comparison_failures.csv was not supplied).

Analysis Availability

analyze.py has been run; table1_effectiveness/table2_test_translation/figure7_ablation/figure8_cost_tool

Protocol and revision provenance

Frozen protocol fields: model, agent_timeout_seconds, codeweaver_package_version.
Protocol-consistent: True.
Strictly identical environment metadata: False.
Recorded informational revision drift (retained exactly and treated as a validity threat, not collapsed)
- git_sha: 6 exact value(s): 3c6f1d63b5d9d868e6f9f3207e11619f944e51e8, 58ef7b06dff42ca4227d7998c0df620
- copilot_cli_version: 6 exact value(s): GitHub Copilot CLI 1.0.79-3., GitHub Copilot CLI 1.0.79-4., G

Cross-System Comparison: Design and Tracks

- Tracks:
- fresh GPT-5.6 Sol CodeWeaver runs
 - released-artifact ReCodeAgent/prior replay
 - workbook-only published reference (non-replayed)

CodeWeaver GPT-5.6 Sol is not model-matched to the original systems/models. Released ReCodeAgent/prior o

Released SWE-agent CRUST targets are unavailable and were never fabricated. The optional workbook can ap

Unavailable costs remain unavailable and are never treated as zero.

Cross-System Comparison: Inventory and Completeness

system	tool	repetition	expected	measured	unavailable	error	missing	accounted_missing
---	---	---	---	---	---	---	---	---
codeweaver	crust	0	100	100	0	0	0	measured
codeweaver	oxidizer	0	6	6	0	0	0	measured
codeweaver	alphatrans	0	4	4	0	0	0	measured
codeweaver	skel	0	8	8	0	0	0	measured
codeweaver	crust	1	100	100	0	0	0	measured
codeweaver	oxidizer	1	6	6	0	0	0	measured
codeweaver	alphatrans	1	4	4	0	0	0	measured
codeweaver	skel	1	8	8	0	0	0	measured
codeweaver	crust	2	100	100	0	0	0	measured
codeweaver	oxidizer	2	6	6	0	0	0	measured
codeweaver	alphatrans	2	4	4	0	0	0	measured
codeweaver	skel	2	8	8	0	0	0	measured
recodeagent	crust	0	100	99	0	0	1	missing
recodeagent	oxidizer	0	6	6	0	0	0	measured
recodeagent	alphatrans	0	4	4	0	0	0	measured
recodeagent	skel	0	8	8	0	0	0	measured
prior	crust	0	100	0	100	0	0	missing
prior	oxidizer	0	6	6	0	0	0	measured
prior	alphatrans	0	4	4	0	0	0	measured
prior	skel	0	8	8	0	0	0	measured

Cross-System Comparison: CodeWeaver Three-Repetition All-Tools Summary

metric	n	mean	sample_sd	ci_95_t	status	reason
---	---	---	---	---	---	---
compilation_success	3	1.0000	0.0000	[1.0, 1.0]	measured	
project_pass_all	3	0.5141	0.0049	[0.5019699075430806, 0.5262786800275407]	measured	
validated_test_micro_pass_rate	3	0.4252	0.0061	[0.41019679772302653, 0.4403015411474053]	measured	
validated_test_macro_pass_rate	3	0.6326	0.0049	[0.6204641186508332, 0.6447762184728173]	measured	

Cross-System Comparison: Primary Rep-0 Paired All-Tools Results

metric	n	cw_yes_rca_no_wins	rca_yes_cw_no_losses	cw_wins	rca_losses	ties	delta_percentag
---	---	---	---	---	---	---	---
compilation_success	117	37	0	80	31.6239	0.0000	
project_pass_all	117	27	15	75	10.2564	0.0884	
validated_test_project_rate	117			12	30	75	-8.7477

Cross-System Comparison: CRUST Three-System Overlap

Status: measured; triples: 99.

codeweaver_rep0	recodeagent_replay	swe_agent_workbook	count	swe_agent_track
False	False	False	0	published_reference_non_replayed
False	False	True	0	published_reference_non_replayed
False	True	False	0	published_reference_non_replayed
False	True	True	0	published_reference_non_replayed
True	False	False	21	published_reference_non_replayed
True	False	True	16	published_reference_non_replayed
True	True	False	37	published_reference_non_replayed
True	True	True	25	published_reference_non_replayed

Cross-System Comparison: Measured Cost/Correctness Frontier

Status: unavailable. fewer than two systems have genuinely measured, same-unit actual costs; missing rep

system	track	cost_status	mean_actual_cost	n_cost_projects	cost_missing_projects	correctnes
codeweaver	measured	measured	2537095697245.7627	118	0	measured
recodeagent	released_artifact_replay	unavailable		0	117	measured
prior	released_artifact_replay	unavailable		0	18	measured

Missing or unavailable cost is not zero and is excluded from the measured frontier.